S01 Temperature and Heat Transfer | 2013 Q2

In (a), not many candidates were able to state the correct procedures to be done before taking a reading. They mentioned stirring the water but not stopping the heat. In (b)(i), many candidates managed to work out the correct numerical answer but some of them used incorrect units. In (b)(ii), the concept of absolute zero by experiment (· = 0) was not well interpreted by some candidates. Many of them just stated ·273°C as absolute zero without deducing the value from the data given.

S01 Temperature and Heat Transfer | 2022 Q1

This question examined candidates' knowledge and understanding of heat transfer. The performance was fair. Most candidates knew the meaning of 'fair test' in answering (a). In (b), although candidates mentioned that the rate of energy loss decreased with time, not many were able to explicitly point out that it was due to the decrease in temperature difference. Part (c)(i) was well answered. In (c)(ii), quite a number of the candidates did not know that a shiny surface is a good reflector/poor radiator of thermal energy. Weaker candidates gave wrong explanations in terms of conduction, for example, aluminium is a good thermal conductor. Part (d) was well answered.